THE UNIVERSITY OF MELBOURNE

Student number

Assignment 8 Semester 1, 2023

School of Mathematics and Statistics

MAST10012 Introduction to Mathematics

Submission deadline: 5:00 PM (17:00, AEST), Thursday 11 May 2023

This assignment consists of 5 pages (including this page)

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Instructions to Students

Writing

- If you have a printer (or can access one), print the assignment one-sided. Write your answers in the boxes provided on the assignment that you have printed.
- If you do not have a printer, and you know how to annotate a PDF using a tablet, then annotate your answers directly onto the assignment PDF and save a copy for submission.
- If you cannot use the template, write your answers on blank A4 paper. Mimic the template by numbering your pages, and note which answers to write on each page.
- Record your student ID number in the top right corner of this cover page (or the first page of your submission), and your details at the bottom of this page (or the first page).
- Provide complete and fully justified solutions for each question.
- Write legibly, and use appropriate notation. Marks will not be awarded to unclear work.
- If you run out of space for a question, append additional work to the end of your submission. Make a note ("See page X") in the relevant question box warning the marker.
- Teaching staff cannot help you directly with assignment questions. Please ask on the Ed Discussion board if you need clarification on the wording of a question.

Scanning

- Put the pages in number order and the correct way up. Add any extra pages to the end. Use a scanning app to scan all pages to PDF. Do not scan on an angle. Crop images to the page borders. Make sure that your PDF file is readable. Material that cannot be read will not be marked.

Submitting

- Go to the Gradescope window. Choose the relevant Canvas assignment. Make sure that you upload the correct PDF file. You should get a Gradescope confirmation email.
- Assignments submitted late will incur a penalty of 4% (of the total mark available) for every hour or part thereof up to 24 hours. Written assignments cannot be submitted more than one day (24 hours) late.

Student Details

- You must provide your student identification (ID) number in the box at the top of this page.
- Do not record your name and tutorial details on this paper. Only your student identification (ID) number is required.

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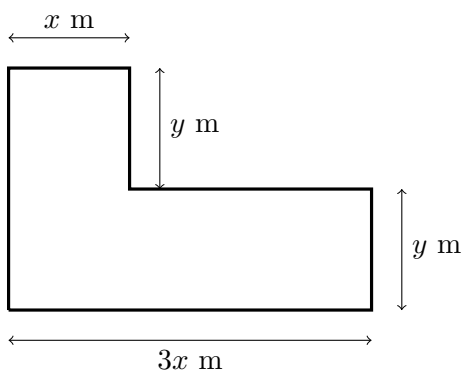
Question 1

A flower bed is L-shaped, as shown in the figure. Its perimeter is $P = 32$ m.



(a) Write an expression for the area $A(x, y)$ m^2 in terms of x and y. Show all your working.

(b) Find y in terms of x.

(c) Calculate an expression for the area, $A(x, y) = A(x)$, in terms of x only.

(d) What is the maximal domain of $A(x)$?

(e) Find the values of x and y which give the maximum area, and state the value of the maximum area. If your answer is fractional, do not express it as a rounded decimal: express it as a fraction.

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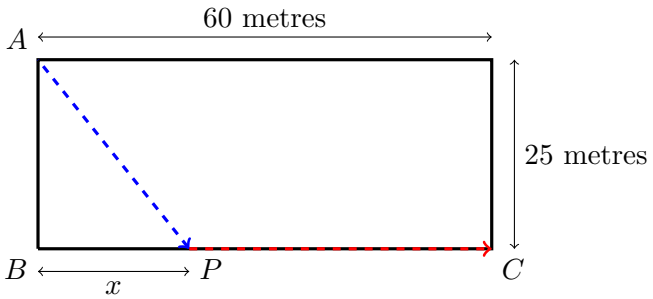
Question 2

A resort at the University of Melbourne has a rectangular swimming pool as illustrated:



Will can swim at a velocity of $v(\text{swim}) = 1$ m/s and run at a velocity of $v(\text{run}) = 5$ m/s. They start at point A, swim to a point P (where P is a point on the line from B to C); they take 3 seconds to pull themselves out of the pool; then run from point P to point C: $A \to P \to C$. Let the distance from B to P be $d(B \to P) = x$ m. Let the total time it takes the student to get from A to C via P be $t(A \to P \to C)$ seconds.

(a) Show that:

$$t(A \to P \to C) = \sqrt{x^2 + 625} - \frac{x}{5} + 15 \text{ seconds}$$

You must clearly explain your reasoning, and show all steps in your working.
This question was previously incorrect. The correct number, 625, is highlighted in red.

(b) Suppose the pool's volume is given by $V(x) = 1500x(x+1)$ m^3, where x is a measure of the pool depth in metres. Water is added to the pool via a hose at a consistent rate of 5 m^3/h. When $x = 1$ m, what is the rate of change of x with respect to time?

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End of Assignment